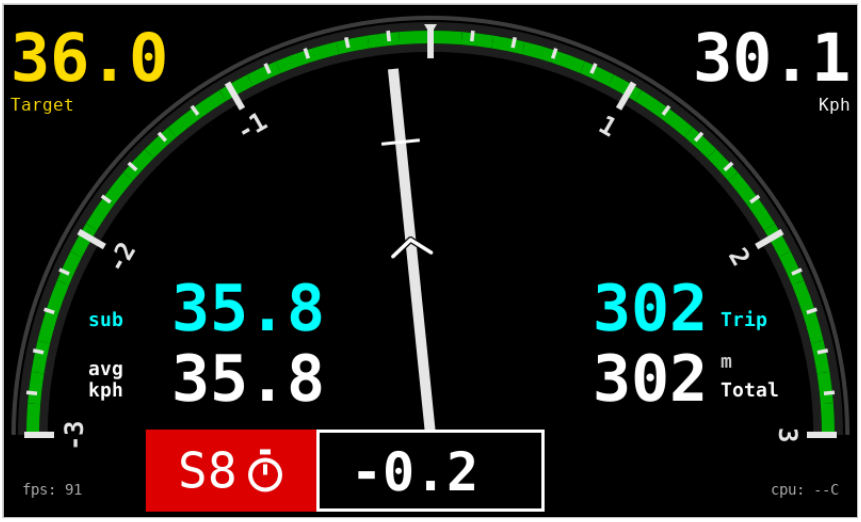
Rallybox User Manual

Software Nero2.1 · manual updated 4 November 2026

Driver Display



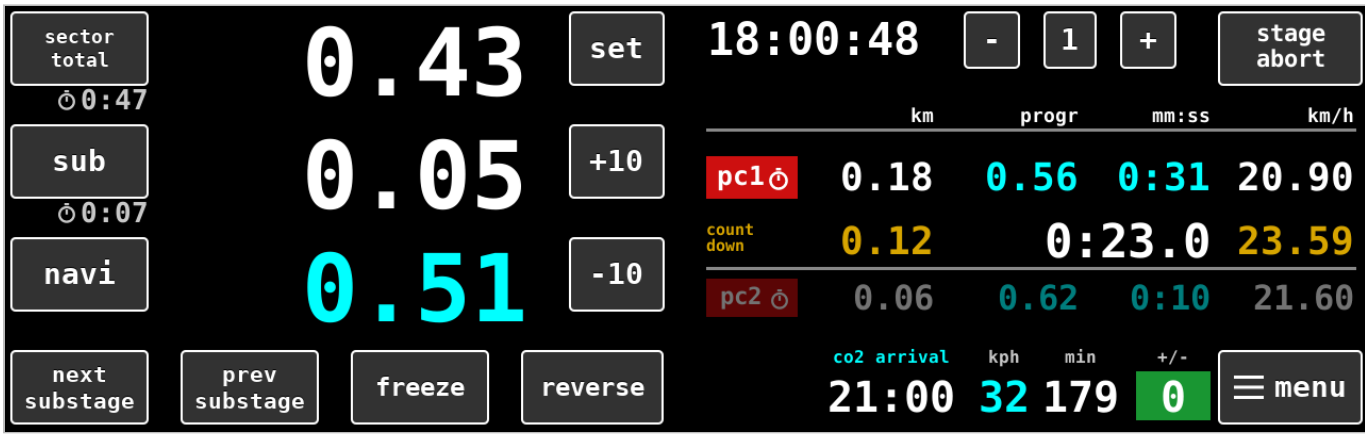
Driver display.

Navi page (greek rally)



Co-pilot display, navigation page (greek rally type, stage running).

Navi page (italian rally)



Co-pilot display, navigation page (italian rally type, inside a timed trial).

CONTENTS

Rally types **3**

Getting around **4**

Greek rally type **5**

- Programming Greek Stages 5
- Starting a Greek Stage 7
- Navigating a Greek Stage 9

Italian rally type **12**

- Programming Italian Sectors 12
- Starting an Italian Sector 14
- Navigating an Italian Sector 15

Surrey rally type **17**

- Programming Surrey Stages 17
- Starting a Surrey Stage 18
- Navigating a Surrey Stage 18

Driver displays **19**

- Greek and Surrey driver display 19
- Italian driver display 21

Single-display mode **22**

The phone app **23**

Beep assist **26**

Setting rally configurations **27**

- DATE / TIME 27
- CALIBRATION 28
- SETTINGS 29
- SETUP 30

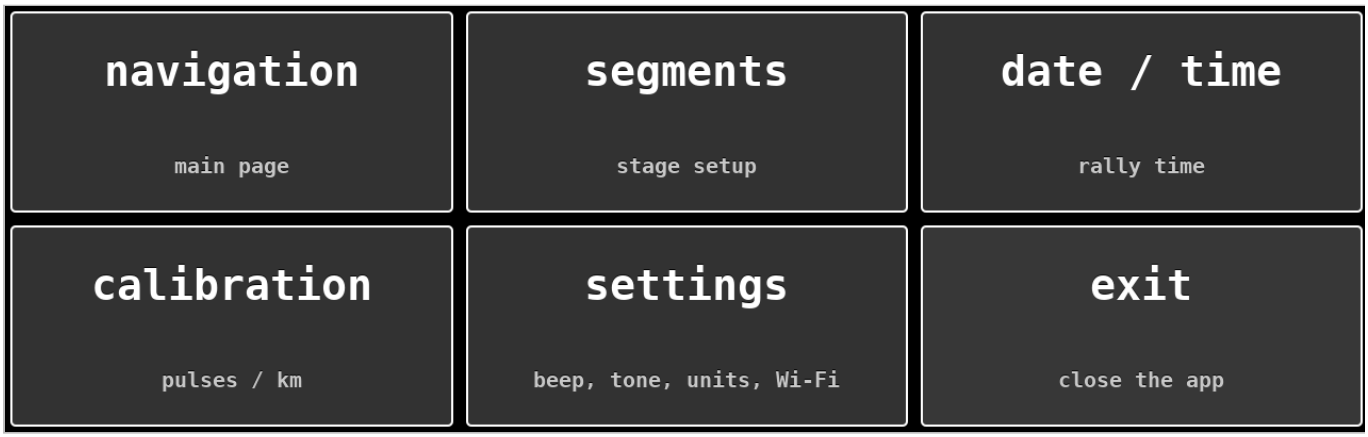
Index **31**

RALLY TYPES

The box has been designed to work with the rallies having various characteristics and operates slightly differently depending on the type of rally selected.

Rally type	Characteristics	In this manual
greek	• Stages start at roadbook boxes. • Distance is zeroed at each stage start. • Departure is at a set minute. • Speeds change at roadbook boxes.	Programming, Starting, Navigating
italian	• A stage covers an entire **sector** from one time control (CO) to the next. • A sector is made up of substages involving: ◦ untimed driving ◦ timed trials (pc) ◦ average trials (pm) ◦ stamp checks (ct) ◦ the closing time control (co) • Sectors start and finish on specific minutes but the intervening substages do not have to start at a specific time. • Total distance is not zeroed at the start.	Programming, Starting, Navigating
surrey	• Stages start at a position (landmark) within a specific roadbook box. • The exact starting distance may be unspecified. • Total distance is not zeroed at the start. • Scored on time taken.	Programming, Starting, Navigating

GETTING AROUND



The menu page.

Every page has a `menu` key at the bottom right. Press it to reach the main menu.

- **navigation** – used while navigating and on stages.
- **segments** – stage setup.
- **date / time** – set rally time.
- **calibration** – set pulses per kilometre.
- **settings** – beep, tone, rally type, units and logging; the spanner key opens the **Setup** sub menu (displays, Bluetooth speaker, Wi-Fi and the phone link, software updates).
- **exit** – saves the settings and closes the program. It does not switch the Raspberry Pi off.

Numbers are typed on the on-screen keypad: `C` clears a selected entry and the arrow key deletes one character. There is no Enter key: an entry is saved automatically when you tap another box or after no activity for a second.

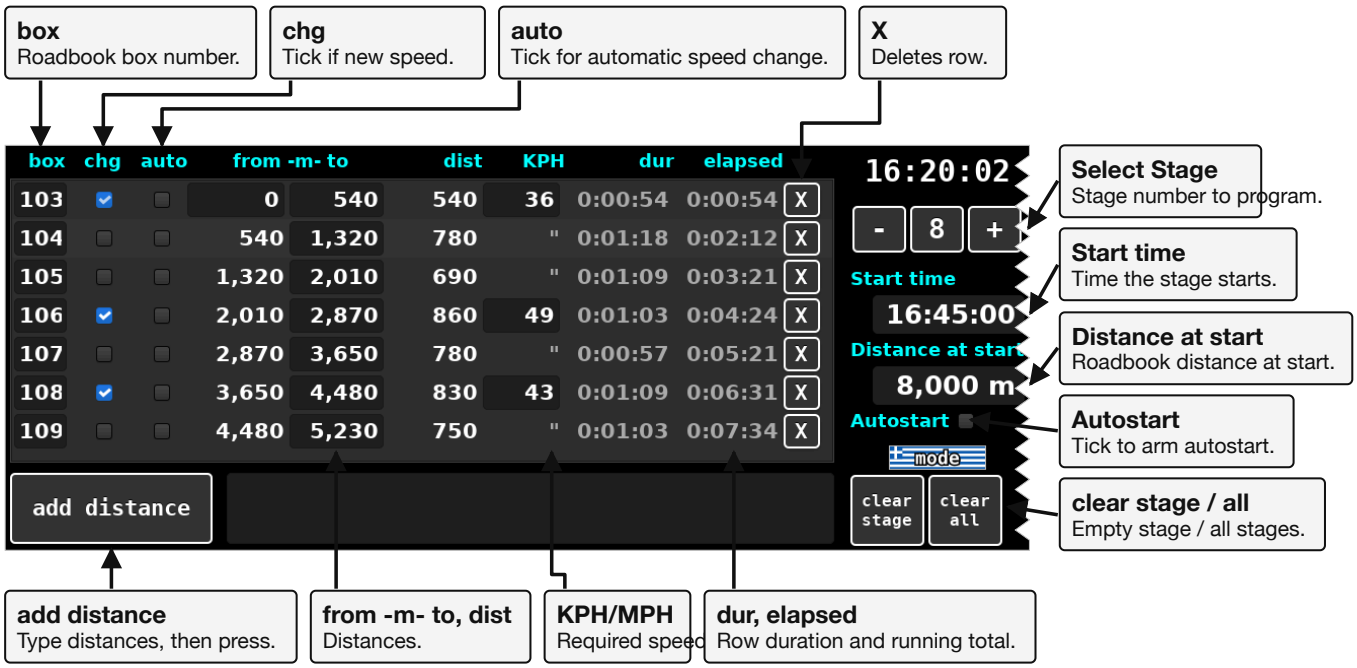
GREEK RALLY TYPE

Characteristics of greek rallies

- Stages start at roadbook boxes.
- Distance is zeroed at each stage start.
- Departure is at a set minute.
- Speeds change at roadbook boxes.

Programming Greek Stages

Select **greek** rally type from the settings page and then select `segments` from the main menu.
The segments page will display a Greek flag.
The system defaults to metres but can be set to KM in settings.



Stage 8: one row per roadbook box, speed changes at boxes 103, 106 and 108.

Entering Greek stage data

When entering stage data you can choose to enter every roadbook box in the stage, or only those rows with speed changes. With every box entered, Beep Assist navigation (when on) beeps ahead of each box.

1. Select the stage with **– n +**, or tap the number (1–99).
2. Tap the box beside `add distance` and type the cumulative distance at which each row ends, separated by semicolons: `540;1320;2010` (or `0.54;1.32;2.01` in km). Later entries are appended or inserted as needed.
3. Press `add distance` and the distances will be added to the summary table.
4. For rows with speed changes tick **chg** and enter the speed.
5. Tick **auto** for the box to change speed automatically at the specified distance. If unticked, the navigator must press next speed on the navigation page.
6. **Start time** – enter when the stage starts (hh:mm:ss).
7. **Distance at start** – enter the roadbook distance at the start line. (If entered the box will show the speed and time needed to arrive at the next stage on time.
8. **Autostart** – tick to arm an autostart at today's start time (a past time is refused). It can also be armed/disarmed on the navigation page.

Entries save automatically under the stage number shown. A running stage cannot be edited.

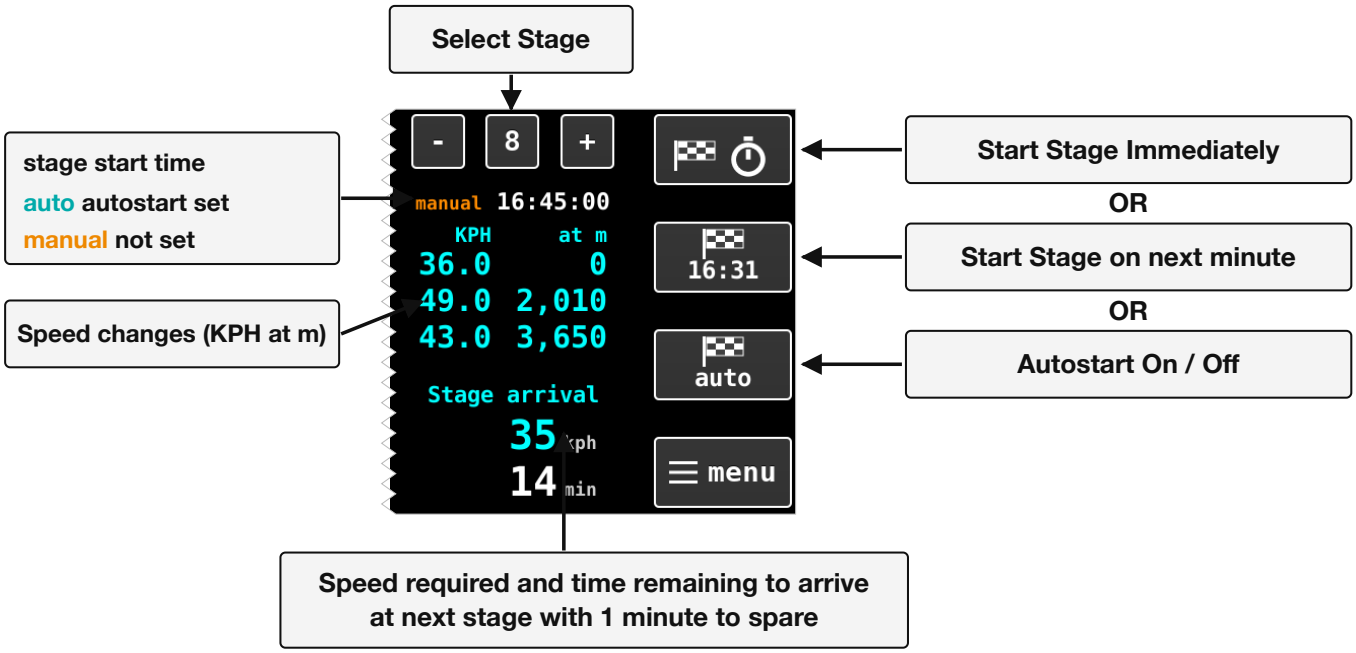
Uploading stages

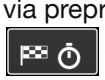
upload stage on the phone's Segment page opens a window showing the CSV columns required for greek, surrey and italian rallies. Choose the CSV, or paste it into the box, then press check and load. If a stage is already programmed it asks for confirmation before it is replaced. The same key on the box opens a file window and loads that file. It is greyed while a stage is running.

Clearing stages

clear stage empties the stage shown; clear all empties every stage. Both ask for confirmation first.

Starting a Greek Stage



A stage can be started via preprogrammed autostart (using the time programmed on the Segments page), or by pressing the Start Now  or Start On Next Minute  buttons.

Autostart: When autostart is armed the button  will appear as black on white, and the word **auto** (cyan) appears next to the start time. When Autostart is not armed the word **manual** (orange) appears. A start time that has passed, with nothing armed or running, is crossed out in red. Autostart can be toggled on/off using the Autostart button on the navigation display.

What happens with autostart:

- With less than 5 minutes to the start the driver's display will show a countdown clock.
- The Total and Trip distances will zero at the start time.
- The timer resets at the start time.

Start on Next Minute: The Start on Next Minute button constantly updates to show the next whole minute. Pressing it sets a type of autostart at that minute.

What happens with Start on Next Minute:

- Pressing the button automatically overrides any programmed autostart.
- The Total and Trip distances zero immediately (not at the departure minute). Any distance driven between pressing and the start minute will be included in the stage distance.
- The timer resets at the start minute.

Start Now: Pressing it starts a stage immediately.

What happens with Start Now:

- Pressing the button overrides any programmed autostart.
- The Total and Trip distances zero immediately.
- The timer resets immediately.
- On a Greek stage the Trip history clears. Italian and Surrey keep it.

Starting scenarios

Regular on-schedule start – in most instances this will be handled correctly by a pre-programmed autostart which resets the distances and timer at the allotted start time.

Late arrival at start – there are 2 ways to handle this scenario.

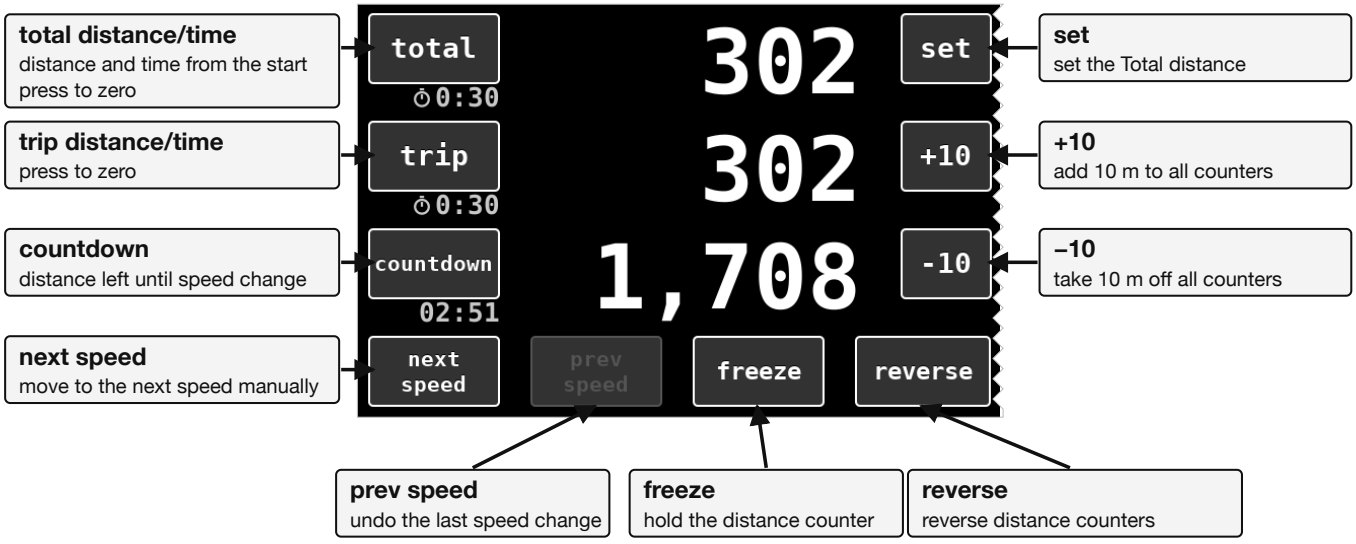
- Have autostart start the stage on time (before you have arrived) and press Total to zero the distances at the start line and attempt to catch up during the stage. This is the preferred method.
- Press Start Now at the line – if a stage is already running the box will ask you to confirm but will measure as of the original button press. Because the timer will have started late the driver will need to catch up and target a reading where the seconds ahead is equal to the delay in starting.

Obstructed start – if traffic or road conditions may make you late to the first timing point, press the next-minute start key (flag with a time) on the start line instead of using autostart, and move off shortly before departure. Total and Trip reset when pressed; timing starts at the minute.

Navigating a Greek Stage



Stage 8 running, 302 m in: 1,708 m and 2 min 51 s to the speed change at 2,010 m. Stage arrival shows the arrival time at the next programmed stage.



total

The total button controls the Distance and elapsed time since the start. Press to zero the distance.

- Outside a stage: pressing zeroes the distance and timer, not Trip.
- Waiting for a timed autostart: pressing zeroes Total, Trip and the timer without asking; the autostart zeroes them again at its start time. (An autostart cannot be armed during a stage.)
- If Start on Next Minute has been selected: pressing zeroes Total and Trip without asking; the timer starts at the minute.
- During a stage (such as after a missed start) pressing to reset the total requires confirmation, but then zeroes Total and Trip from the time of the first press; the stage clock keeps running.

trip

Shows intermediate distance and time. Press to zero it. Prior figures appear in the **Trip history** column. Trip also restarts at every speed change. Starting a Greek stage clears the trip.

countdown

The distance and time to the next speed change are displayed next to the button. The button cannot be pressed.

next speed and prev speed

`next speed` is used to move to the next programmed speed manually. The current speed is displayed in white in the stage panel. `prev speed` undoes the last speed change and the box recalculates the data as if the speed change had not taken place. Both buttons override previous manual and automated speed changes.

+10, set and –10

`+10` / `–10` increment all the distance counters 10 m per press; `–10` cannot take a counter below zero; `set` allows a precise distance to be entered for Total (and Trip remains unchanged). During a stage, corrections apply across average speeds and timing errors.

freeze and reverse

Pressing `freeze` stops the distances from incrementing; pressing freeze again starts the distance incrementing from the frozen figures. Clocks and timers continue to increment while frozen – use it after a navigation error.

`reverse` makes Total and Trip (as well as countdown) increment backwards. It is typically used when reversing after overshooting a junction as the box cannot determine the direction of travel. With reverse pressed the distances can become negative.

Freeze overrules Reverse if both are pressed. The buttons display as black on white when active and a whooping sound is emitted.

A-B, B-C zero

Where a rally is measured on an AB, BC basis and this option has been selected in settings, the `A–B, B–C zero` button appears under the Trip history in the navigation window. Pressing this during a stage when a timing mark or timing marshal is seen sets the timing error in the driver's display to zero and the average speed calculation is made independently between each timing mark.

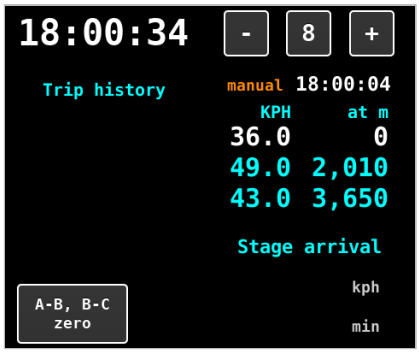
The popup shows the current error size to be removed and the adjustment already applied.



- `Yes` zeroes the needle at this instant.
- `Reset to 0.0` removes the adjustment so the true error shows.
- `No` does not make and changes and returns the box to the navigation screen.

The stage panel display

The stage panel shows the selected stage, the start time of the stage and autostart status, the distances at which speed changes commence (the current segment is shown in white) and the speed needed (and allotted time) to reach the next stage with 1 minute to spare.



The stage panel: stage 8 selected, start time and autostart status, and the speed segments (current segment white).

ITALIAN RALLY TYPE

Characteristics of italian rallies

- A stage covers an entire **sector** from one time control (CO) to the next.
- A sector is made up of substages involving:
  - untimed driving
  - timed trials (pc)
  - average trials (pm)
  - stamp checks (ct)
  - the closing time control (co)
- Sectors start and finish on specific minutes but the intervening substages do not have to start at a specific time.
- Total distance is not zeroed at the start.

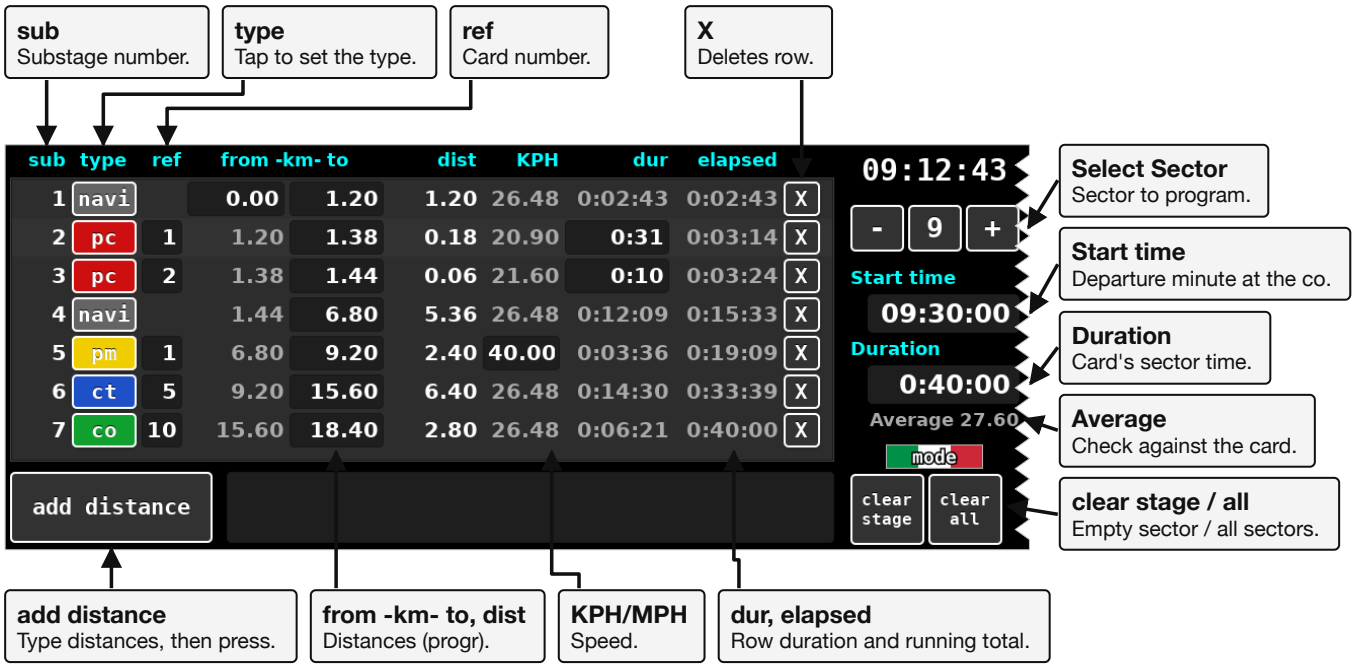
Italian rallies have many short back-to-back timed trials without a pause between them so each time-control section is programmed as linked substages.

The substage types:

Type	What it is	How the box treats it
navi (grey)	Untimed driving between controls.	A suggested speed is worked out from the sector's total duration on order that the car arrives at the final time control in the correct mminute
pc (red, stopwatch)	A imed trial between two pressure tubes.	Speed is worked out from the subsector's duration (not distance).
pm (yellow)	An average speed check with unknown timing points.	Duration is worked out from the given speed (not the elapsed time).
ct (blue)	A substage ending with a Stamp check	Untimed, like navi.
co (green)	A substage that ends with a time control ending the sector; scheduled arrival within a minute.	This the last row of the sector. The arrival time is worked out from the card.

Programming Italian Sectors

Select **italian** rally from the settings page and then select `segments` from the main menu. The segments page will display an Italian flag. The system defaults to km (two decimals) but this can be overridden in settings.



An example first sector, co 1 → co 2, with one of each substage type: navi, two linked pcs, navi, pm, ct and the closing co.

Entering Italian sector data

Enter every substage on the organiser's Times and Distance Table (TDT) starting from the row after the opening time control, up to and including the closing time control (here co 2). The opening time control (co) line is not entered where it has zero distance (0.00 km).

1. Select the sector with **– n +**, or tap the number (1–99): the number of its starting time control.
2. Tap the box beside `add distance` and enter the cumulative distances (**progr**) at which each substage ends either one at a time or enter several separated by semicolons. (eg `1.20;1.38;1.44;6.80;9.20;15.60;18.40` (or `1200;1380;1440;6800;9200;15600;18400` in metres)). Subsequent entries are appended or inserted as needed; take care when inserting a distance inside an existing pc or pm which will get recalculated.
3. Press `add distance`.
4. Tap the **type** tag to cycle to the correct type for each row (navi → pc → ct → pm → co). Each new row for a pc, pm, ct or co substage gets allocated an incremental reference number in the row column but this can be overridden if incorrect.
5. For pc rows enter the card's time (m:ss) in **dur**; the speed is worked out.
6. For pm rows enter the speed in the KPH/MPH column; the duration is worked out.
7. **Start time** – enter the departure minute at the opening co (hh:mm:ss).
8. **Duration** – enter the card's sector duration (h:mm:ss). The **Average** displayed faintly below should correspond to that given on the TDT sheet.

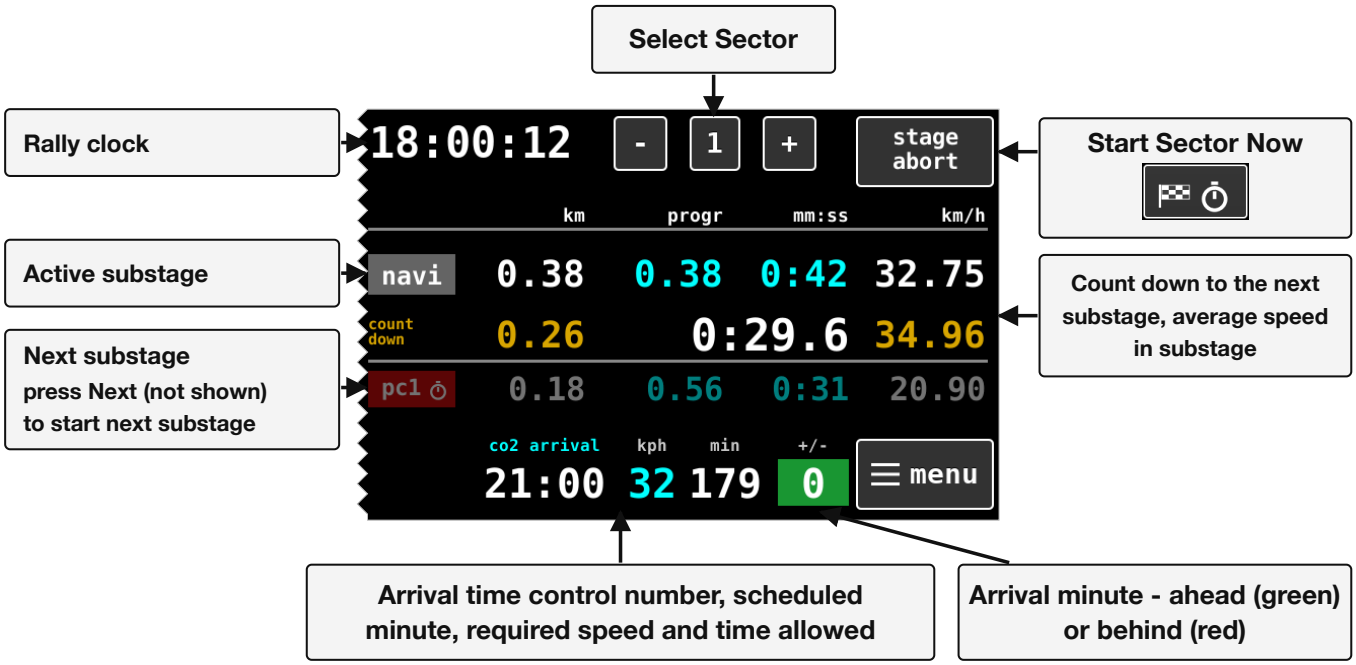
Figures which can be edited are displayed in dark boxes; figures which are calculated and cannot be edited have grey boxes in the segments table. Entries are saved automatically. A running sector cannot be edited.

`upload stage` on the phone's Seg page opens a window. It shows the CSV columns required for greek, surrey and italian stage types, and marks the one this box is using. Choose the CSV, (or paste it into the box if using the phone app), then press `check` and `load`. A stage that is already programmed asks before it is replaced. The button is greyed while a stage is running.

Clearing sectors

`clear stage` empties the sector shown; `clear all` empties every sector. Both ask for confirmation first.

Starting an Italian Sector



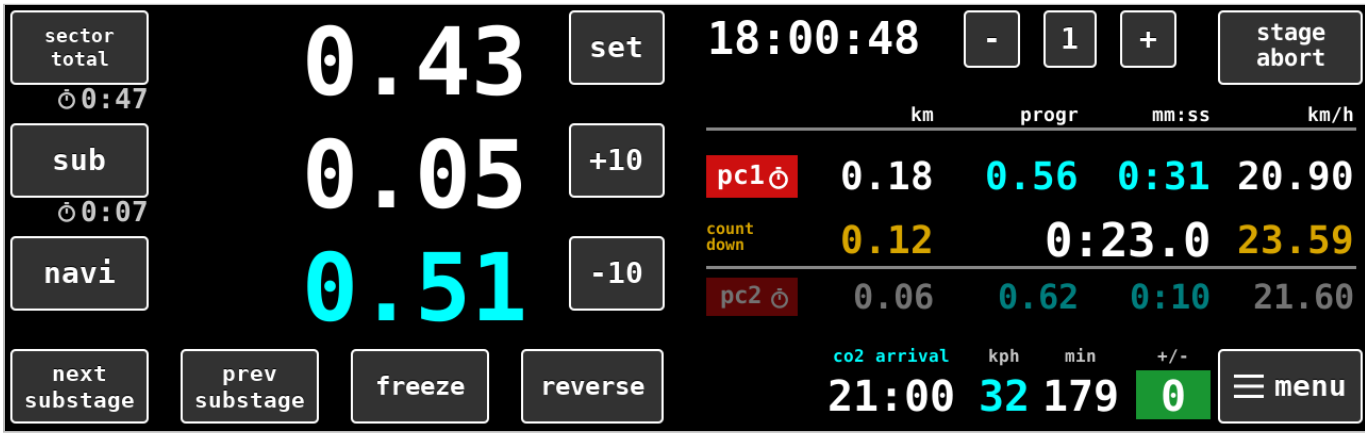
A sector is started at the time control by pressing the start now button during the **departure minute**. The sector clock starts at the second you press, and the arrival target becomes that same second plus the card's sector duration. Sector total and sub distances zero; navi distance does not zero automatically. The `stage now` key reads `stage abort` when a sector is in progress. The box will start in the first substage.

Starting a new substage

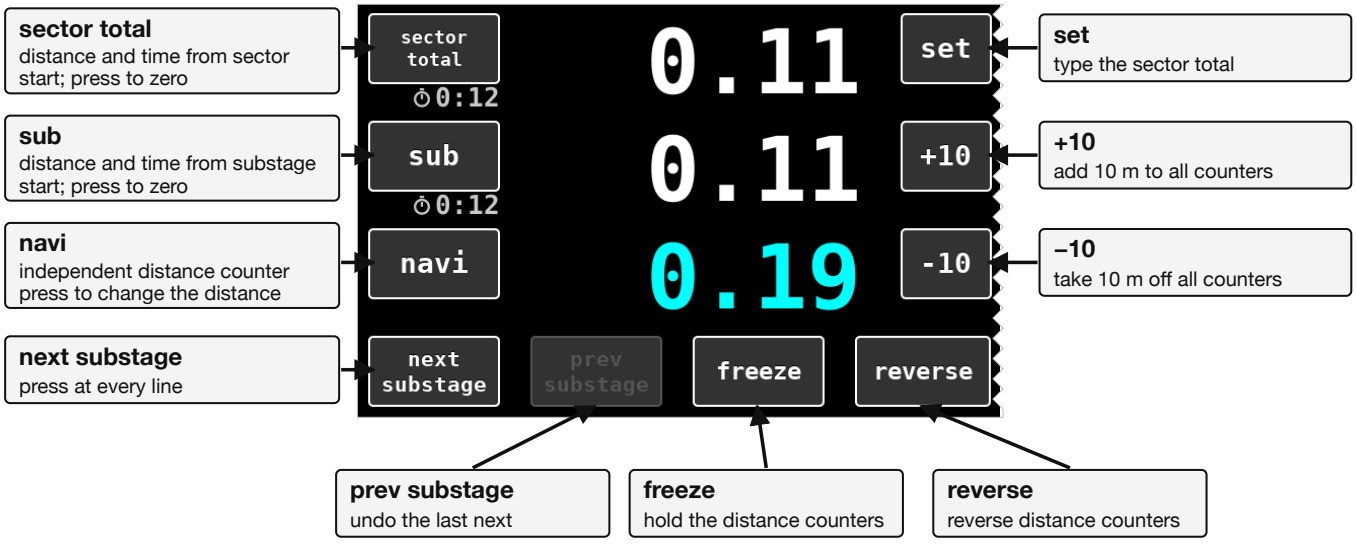
- Press `next substage` to enter the following substage (pc, pm etc). NB that this sets the sector total distance counter to the total distance on the card's **progr** entry. The navi distance increments independently – any variation will give a guide to calibration errors as long as Navi is not reset.
- If `next` substage is pressed early or by mistake press `prev substage` to undo it.

At the final substage.
On arrival at the final time control (co) pressing `next substage` will end the sector. It will warn and ask for confirmation and if confirmed will end the sector and increment the sector selector to the next sector.

Navigating an Italian Sector



Inside pc 1, 60 m in: 120 m and 23.0 s to its end tube; average so far 23.59.



sector total

Distance and elapsed time in the sector, matching the card's **progr** value. Press to zero.

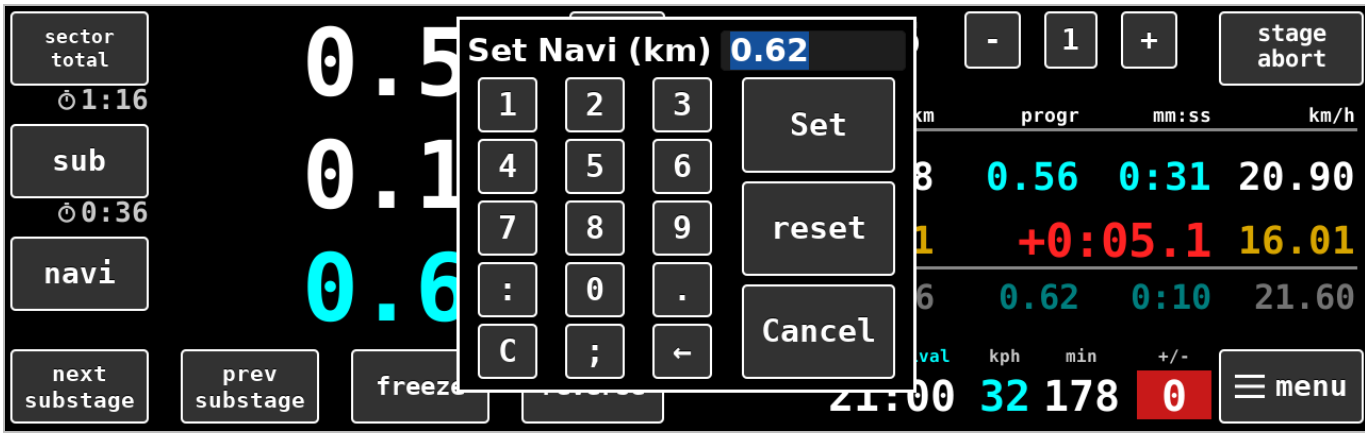
- Outside a sector, it zeroes the distance and the timer but not sub.
- During a sector, it asks first, then zeroes sector total and sub from the press while the sector clock keeps running.

sub

Distance in the current substage; zeroes at every new substage when `next substage` is pressed. Press to zero; inside a pc or pm it asks first (**In stage / Reset sub now?**, with `RESET SUB` or `Back`), as sector total does. The phone asks too.

navi

An independent distance counter, shown in cyan. Press to change the distance. A popup offers the option to reset or specify the distance.



Tapping navi: Set Navi.

next substage and prev substage

`next substage` is pressed to move to the next substage. `prev substage` undoes the change.

+10, set and –10

`+10` / `–10` increment all the distance counters (sector total, sub and navi) 10 m per press; `–10` cannot take a counter below zero; `set` allows a precise distance to be entered for sector total.

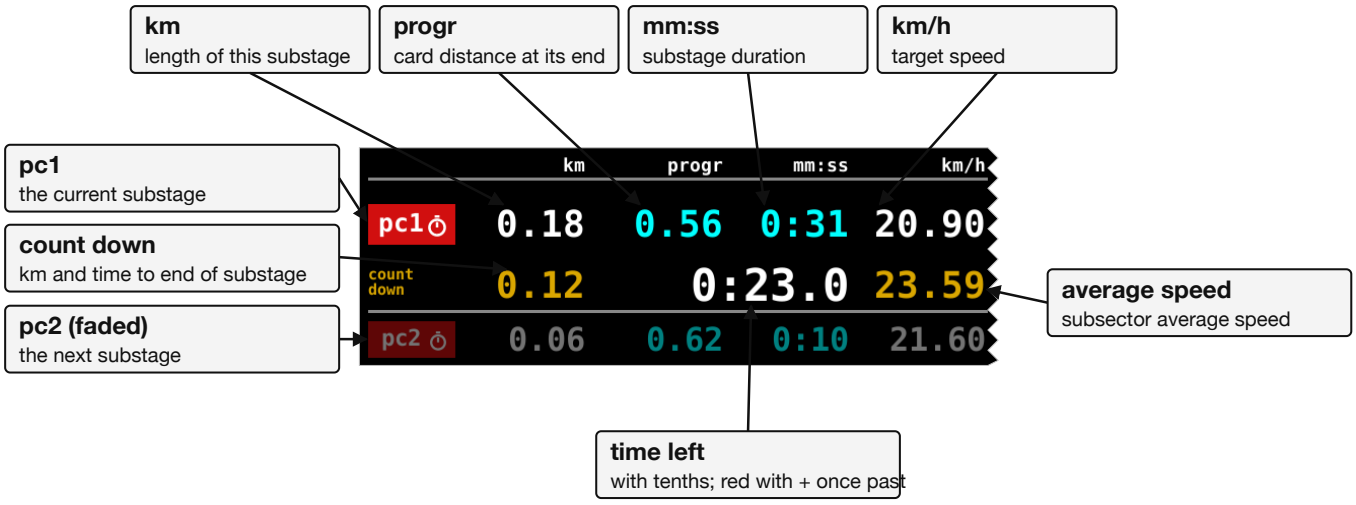
freeze and reverse

Pressing `freeze` stops the distances from incrementing; pressing freeze again starts the distance incrementing from the frozen figures. Clocks and timers continue to increment while frozen – use it after a navigation error.

`reverse` makes Total and Trip (as well as countdown) increment backwards. It is typically used when reversing after overshooting a junction as the box cannot determine the direction of travel. With reverse pressed the distances can become negative.

Freeze overrules Reverse if both are pressed. The buttons display as black on white when active and a whooping sound is emitted.

The substage display



The substage display, inside pc 1.

The count down time to the end of the substage is displayed in large white characters. Past the end of a substage, the count down km (for example `+0.04`) and seconds (for example `+0:01.4`) turn red and count upwards.

SURREY RALLY TYPE

Characteristics of surrey rallies

- Stages start at a position (landmark) within a specific roadbook box.
- The exact starting distance may be unspecified.
- Total distance is not zeroed at the start.
- Scored on time taken.

Surrey works as greek except as noted here.

	greek	**surrey**
Readouts	**total** / **trip** / **countdown**	**total** (from the `now` press) / **trip** / **navi**, the roadbook's running total, set by hand. countdown moves to the stage column.
Stage start	on a time (autostart) or `now`	`now` pressed at the landmark (autostart still works if a start time is entered)
Speed changes	at the row distance, `auto` or `next`	the same, at the roadbook total of the box
Stage end	on distance, or `next` on the last row	`next` pressed at the finish landmark; on distance if not pressed
Result	ahead / behind	ahead / behind while driving, then **time taken** (`9:20.6`)

Programming Surrey Stages

Select **surrey** rally from the settings page and then select `segments` from the main menu. The segments page will display a Union Jack and is otherwise as in Programming Greek Stages.

Entering Surrey stage data

Entry is as per greek except that Start time is usually blank. The absolute distance of the stage start is unknown – enter the distance at the start of that box instead; the end distance may be unknown – enter the distance at the end of the box instead.



The Surrey Segments page, stage SS2: from the start of box 27, speed changes at the end of box 27 and box 30.

`upload stage`, `clear stage` and `clear all` work as in greek.

Starting a Surrey Stage

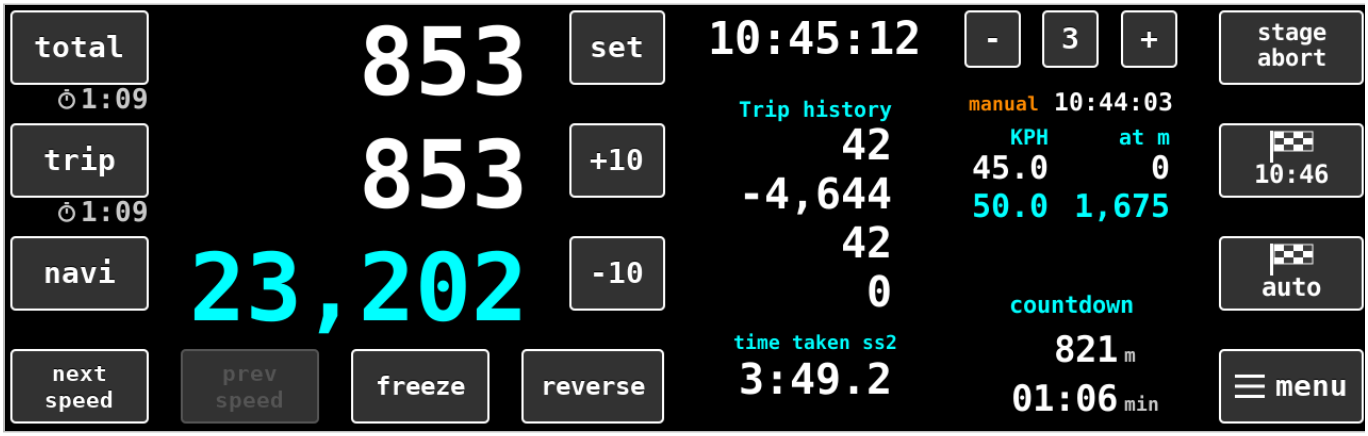
A surrey stage is started with now (Start Now) as the car passes the start landmark. Autostart still works if a start time is entered.

Driving a stage

Set **navi** to the roadbook's figure at each box if it has drifted (navi › set), and always at the start box; reset it to zero where the roadbook restarts (box 1, box 66). Press now as the car passes the start landmark: the stage clock and total start, navi carries on, and row 1 is cut short to end at its box so later speed changes keep to the roadbook. Speed changes then happen by themselves at each box (next and prev work as in greek). At the finish landmark press next on the last row: the stage ends and the **time taken** stays under Trip history and on the phone.

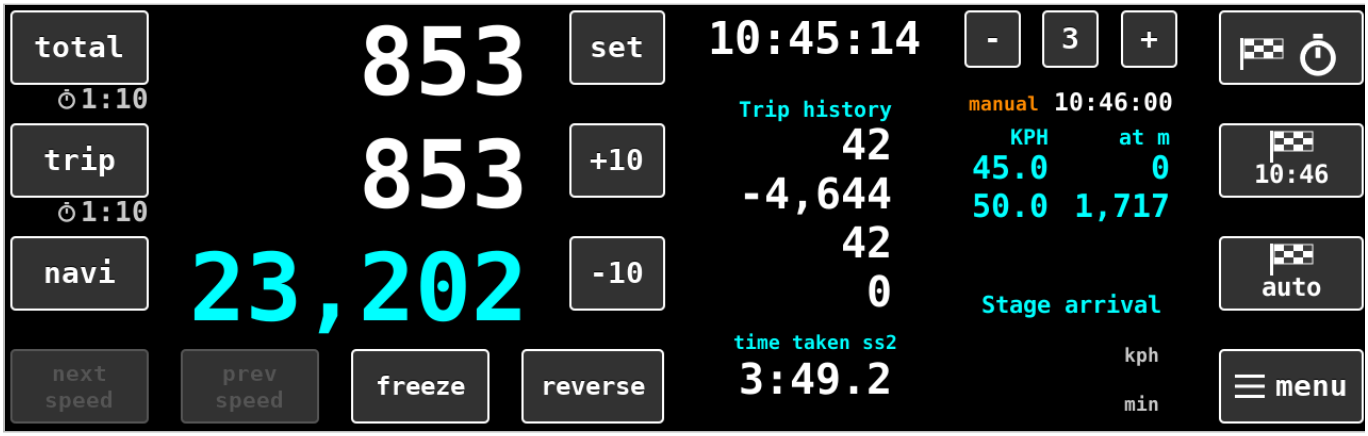
Navigating a Surrey Stage

The navigation page



A Surrey stage running: total from the now press, trip, and navi (cyan) as the roadbook's running total.

Rows: total (from the now press) / trip / navi (cyan, the roadbook running total; tap to set or reset). During a stage, **countdown** (metres and minutes to the next speed change or finish) sits at the foot of the stage column; between stages Stage arrival shows there. Under the Trip history (last four), **time taken** (shown as ss and the stage number) for the last stage stays until the next one finishes. The phone shows navi and time taken. Beeps follow navi, counted from the start box. The countdown and Stage arrival figures each sit above their legend. A-B, B-C zero is available as in greek.



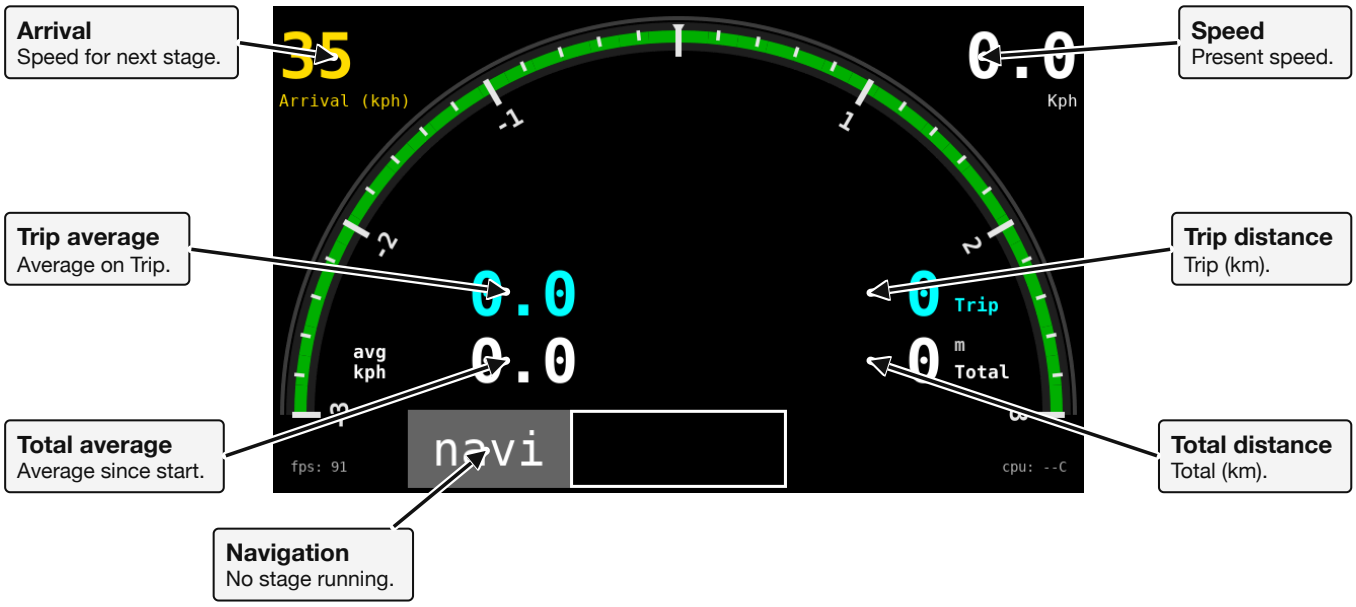
Between stages: the **time taken** for the last stage (ss2, 3:49.2) stays under Trip history.

DRIVER DISPLAYS

Greek and Surrey driver display

The driver display has two designs in Greek/Surrey mode - one for navigation and one in a stage.

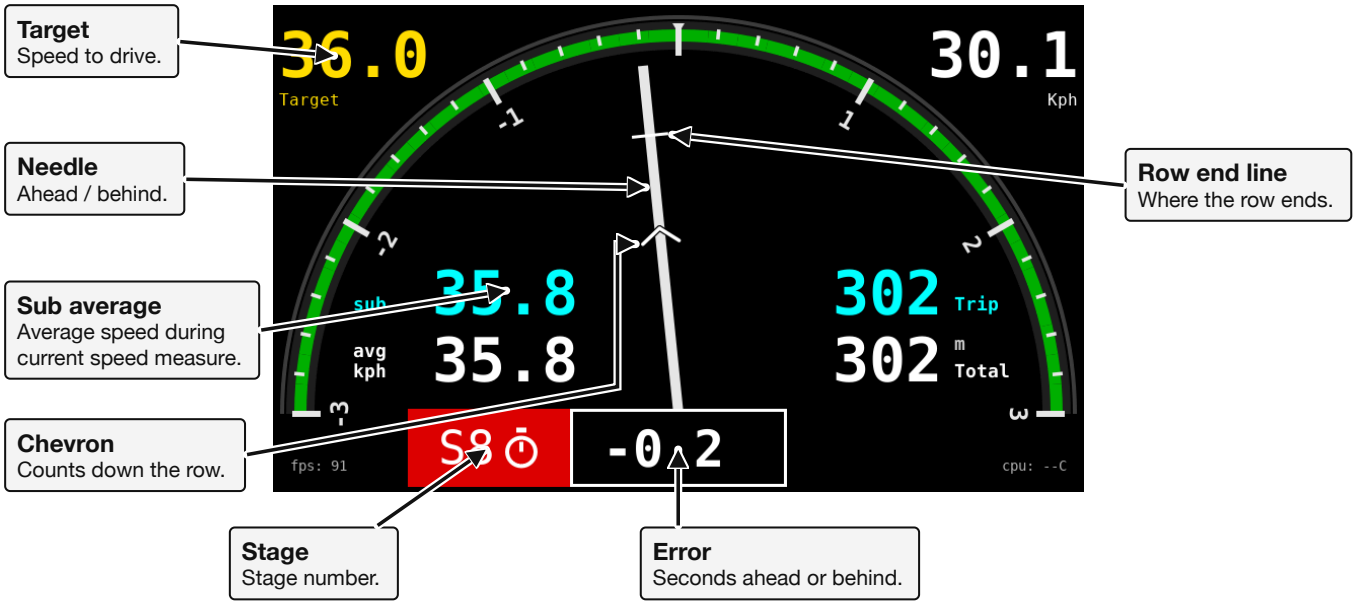
Navigation (between stages)



With no stage running there is no needle. The top left shows the **arrival** speed: the speed needed to reach the next stage with a minute to spare. The distances and averages are Trip and Total.

In a stage

The stage display differs where marked.



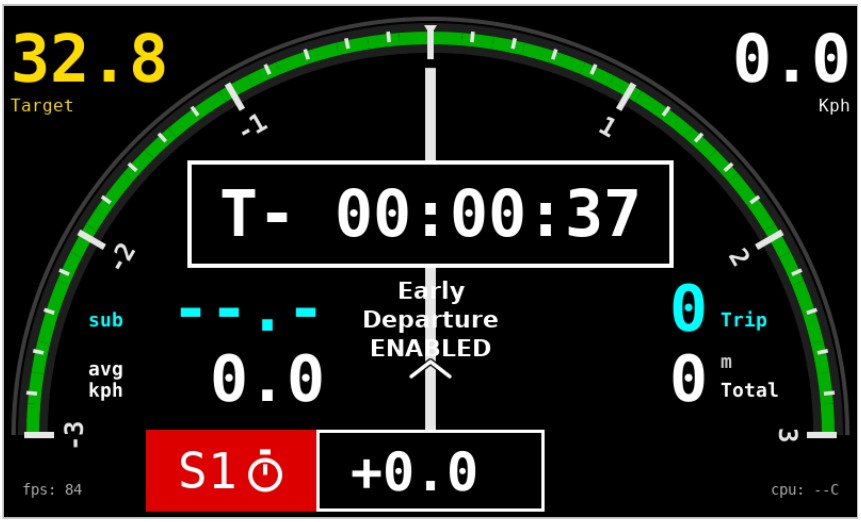
During a timed stage the needle shows how far ahead/behind the driver is in seconds. If the error exceeds the current gauge scale (changeable from 1–3 seconds in settings) the needle will stay pinned far left (or far right if ahead) and the scale will expand. The red logo shows the stage number (S1, S2 …). The cyan **sub** average is your average speed since the last speed change. It restarts at each speed change, so it corresponds with the time error; the white average is the average since the total counter started.

The chevron

The chevron slides up the needle to visually represent the impending end to the current programmed segment row which has been programmed (and may include navigations, speed changes and the finish line). At the start of each row it sits 20% of the way up the needle; as the row is driven it slides up toward the horizontal **row end line**, arriving as the row ends.

Autostart warnings

While an autostart is pending, a T– `hh:mm:ss` countdown shows mid-display with **Early Departure ENABLED** or **DISABLED** beneath, depending on whether a regular autostart or a start on next minute is triggered.

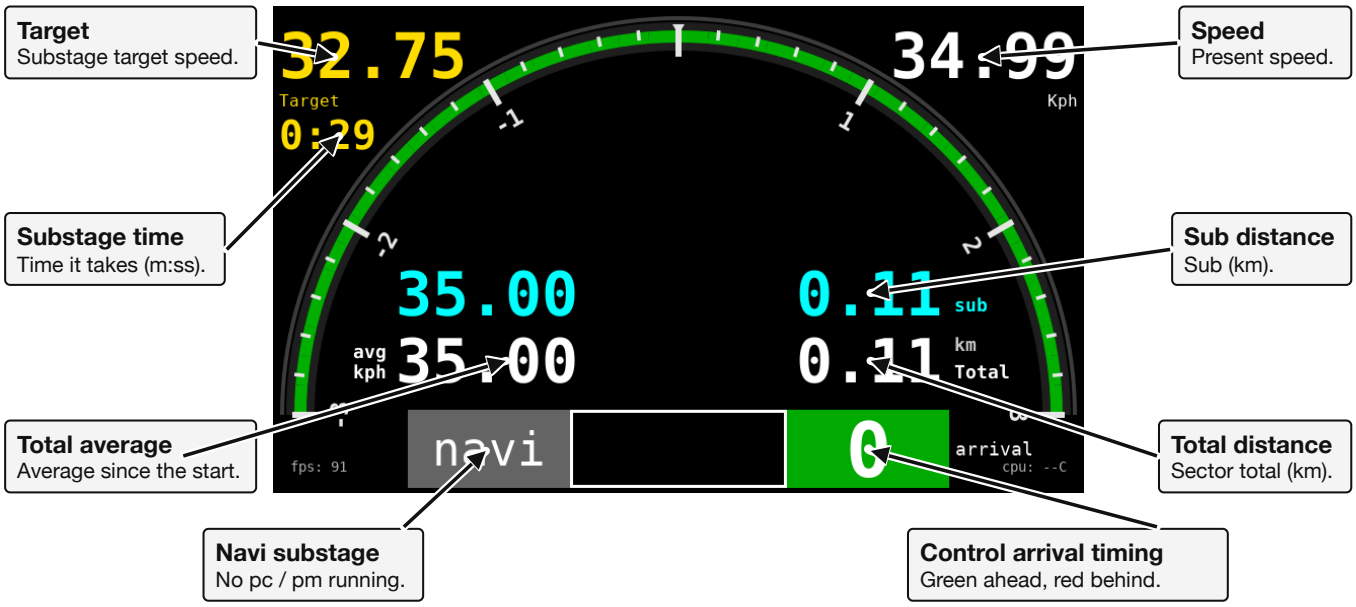


Driver display while a start is pending with Early Departure ENABLED: the coming stage (S1) and its target speed, with the T- countdown and Early Departure ENABLED beneath.

Italian driver display

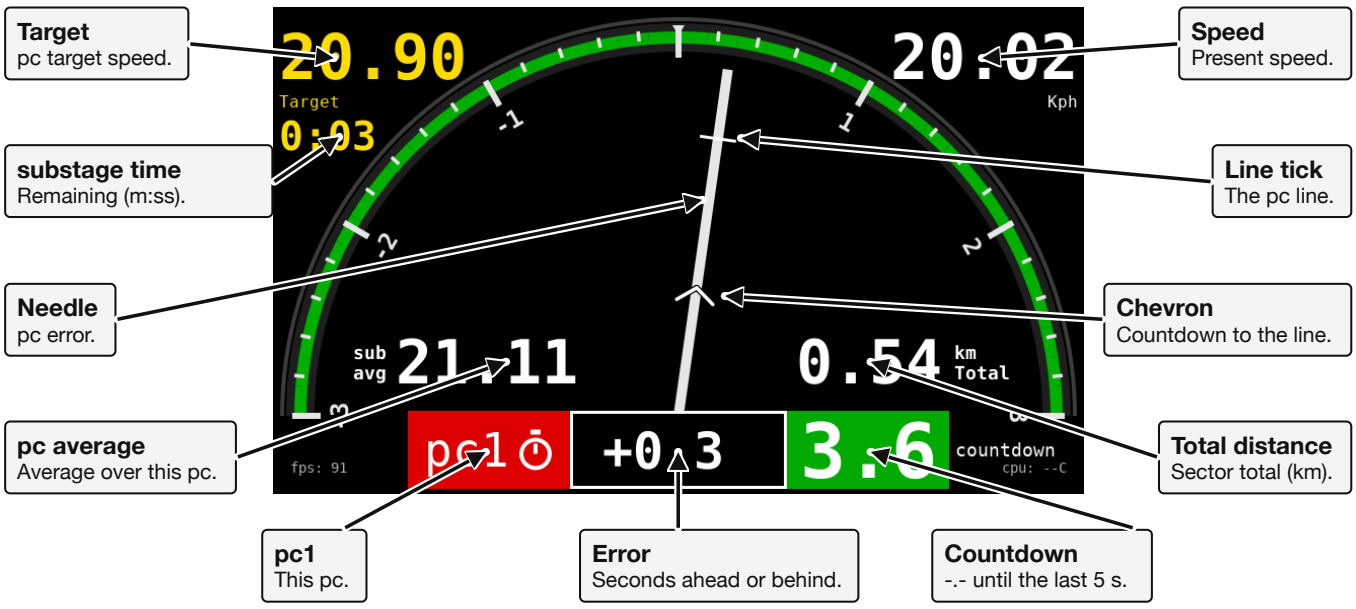
The driver display has two designs in Italian mode - one for untimed/navigation substages and one for timed (pc & pm) substages.

Navigation (navi, ct and co rows)



The display is similar to those for greek/surrey rallies except the right-hand box shows the **minutes** in hand to the control arrival minutes: green when ahead, red when behind.

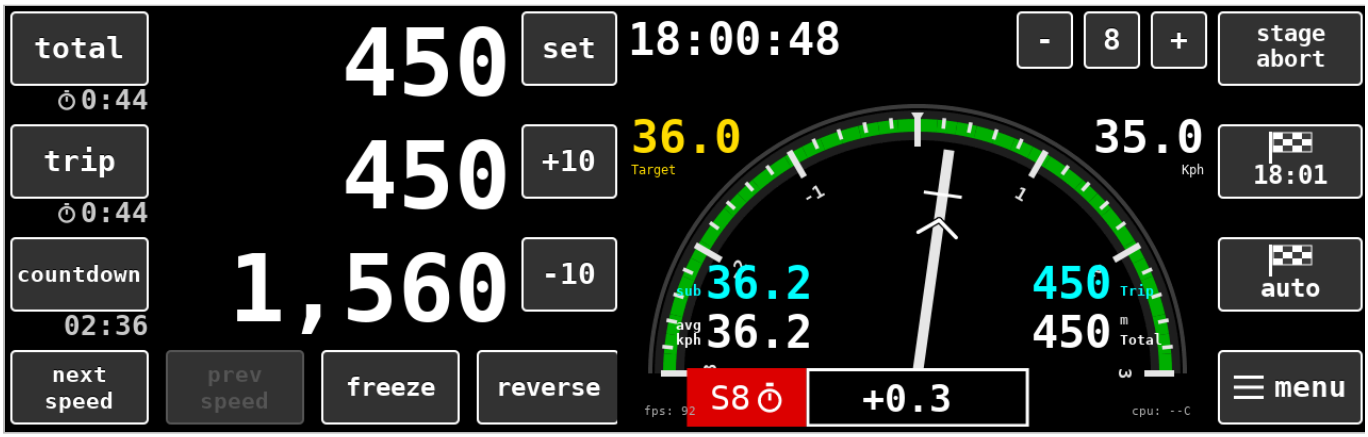
In a pc or pm



The chevron

Due to the short nature of many timed stages, and the increased timing accuracy in Italian stages, the chevron only moves in the last 5 seconds of the stage to visually count down to the end.

SINGLE-DISPLAY MODE



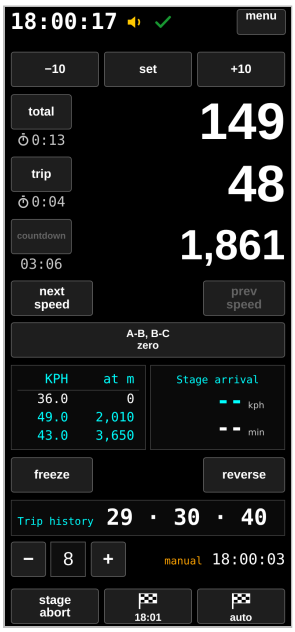
With **Force single display** on, the driver's gauge shares the co-pilot screen. Keys work as on the navigation page. Left: Total / Trip / next with `+10`, `set` and `−10`; along the foot: `prev speed`, `freeze`, `reverse` and `A–B, B–C zero`; middle: rally clock and **– 1 +** stage selector above the gauge; right: the three flag keys and `menu`. No Trip history or stage table; other pages are reached through `menu`. A pending start's countdown shows over the gauge.

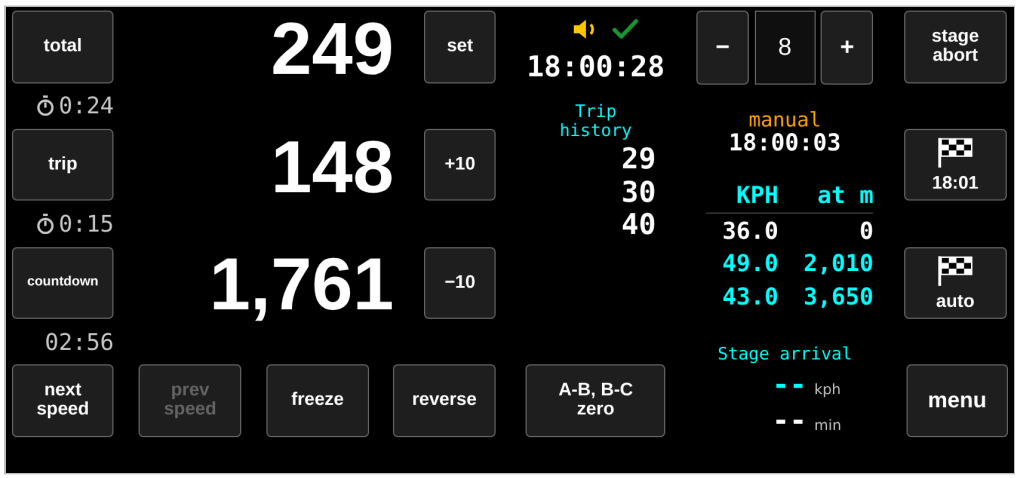
THE PHONE APP

The phone app works within a phone browser window if the box and phone are on the same network. The box or a phone can be set as the hub. The advantage of using the box as the hub is that multiple devices can be used as displays and it has proven reliable in testing. The disadvantage is that connected phones lose their network connection. The advantage of using the phone as the hub is that it can continue to synch its clock and receive messages, however, multiple wireless screens cannot be attached and when using an iPhone the system will not reconnect automatically if the phone ever loses reception (which is frequent in remote parts).

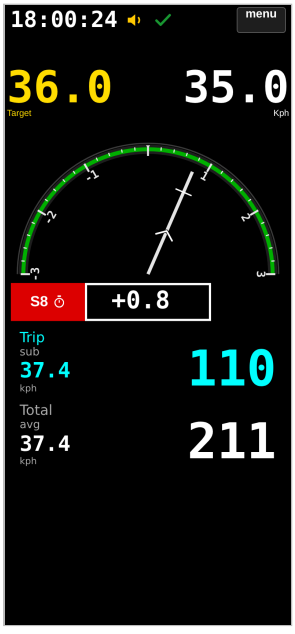
Connect the phone app to the box using the instructions on the settings page and navigate to the displayed browser address. The page will load more cleanly if the web page is saved to the home screen as this disables tabbed browsing. All box features are mirrored on the phone. In landscape views mirror the box display as closely as possible; in portrait the displays are redesigned to fit the screen.

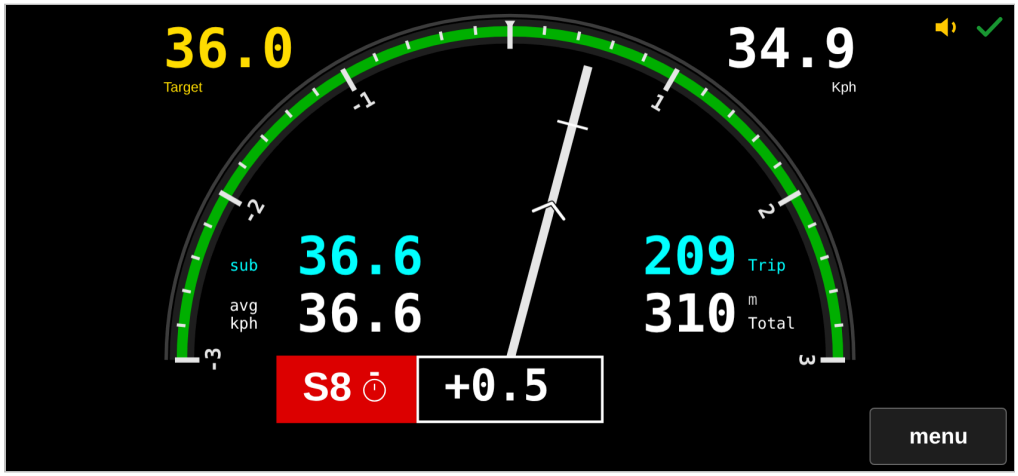
Navi





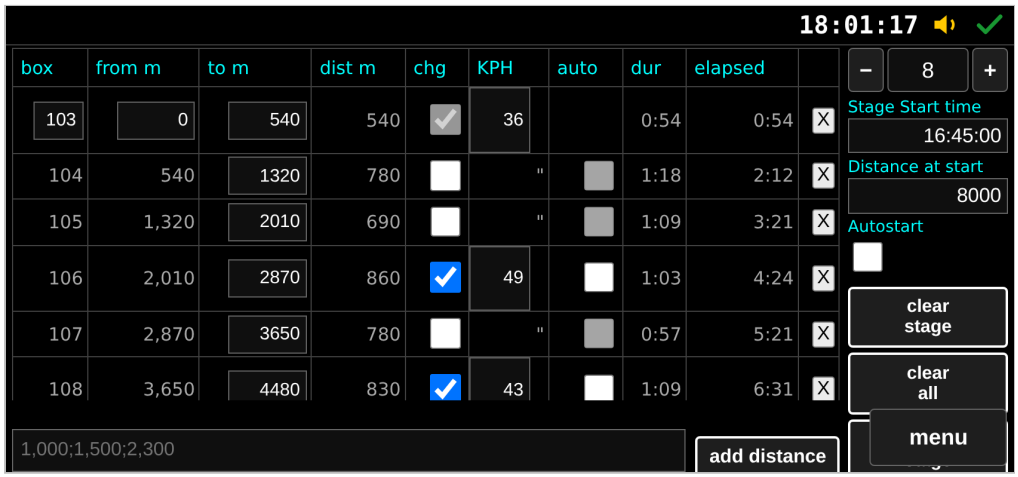
Drv



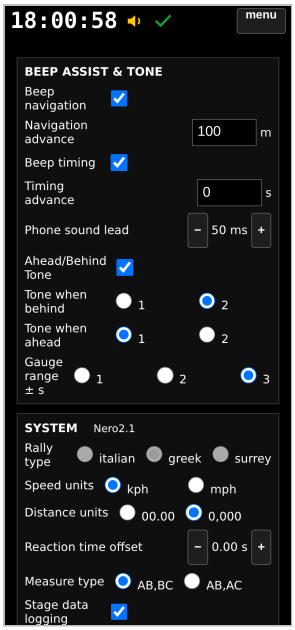


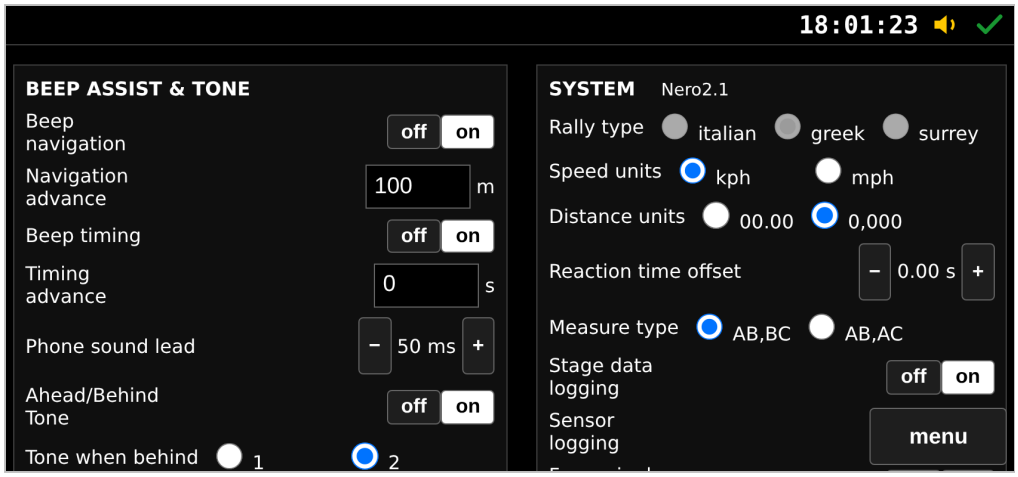
Seg


18:00:52
menu
- 8 +
box
from m
to m
dist m
chg
103 0 540 540
104 540 1320 780
105 1,320 2010 690
106 2,010 2870 860
107 2,870 3650 780
108 3,650 4480 830
109 4,480 5230 750
Stage Start time
16:45:00
Distance at start
8000
Autostart
1,000;1,500;2,300

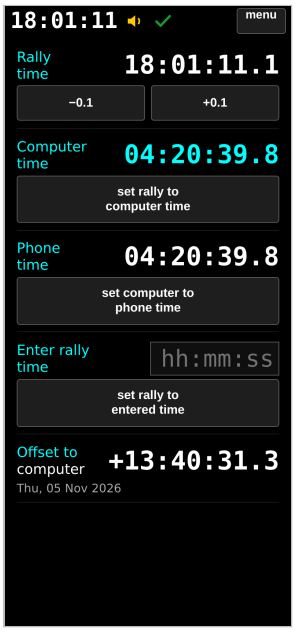

18:01:17
box
from m
to m
dist m
chg
KPH
auto
dur
elapsed
103 0 540 540 36 0:54 0:54
104 540 1320 780 " 1:18 2:12
105 1,320 2010 690 " 1:09 3:21
106 2,010 2870 860 49 1:03 4:24
107 2,870 3650 780 " 0:57 5:21
108 3,650 4480 830 43 1:09 6:31
- 8 +
Stage Start time
16:45:00
Distance at start
8000
Autostart
clear stage
clear all
menu
1,000;1,500;2,300
add distance

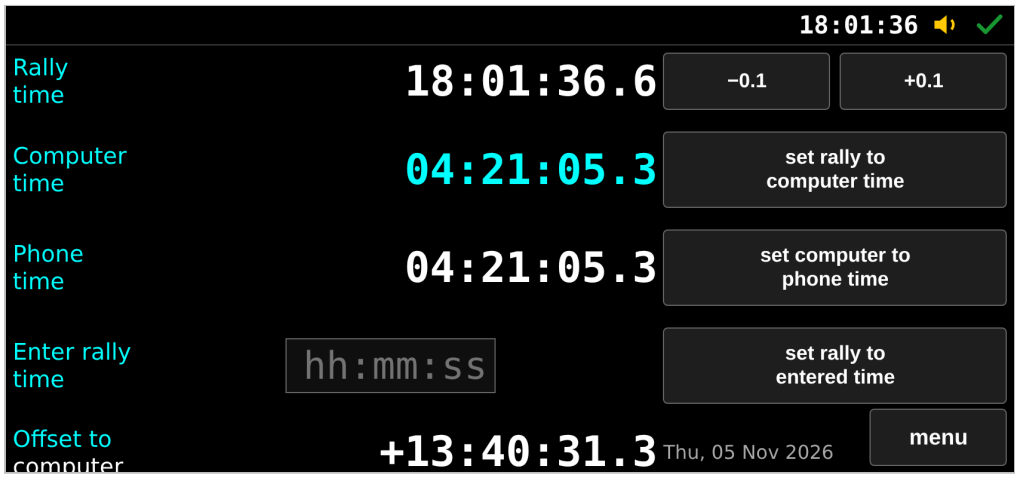
Set


18:00:58
menu
BEEP ASSIST & TONE
Beep navigation
Navigation advance
100 m
Beep timing
Timing advance
0 s
Phone sound lead
- 50 ms +
Ahead/Behind Tone
Tone when behind 1 2
Tone when ahead 1 2
Gauge range ± s 1 2 3
SYSTEM Nero2.1
Rally type italian greek surrey
Speed units kph mph
Distance units 00.00 0,000
Reaction time offset
- 0.00 s +
Measure type AB,BC AB,AC
Stage data logging

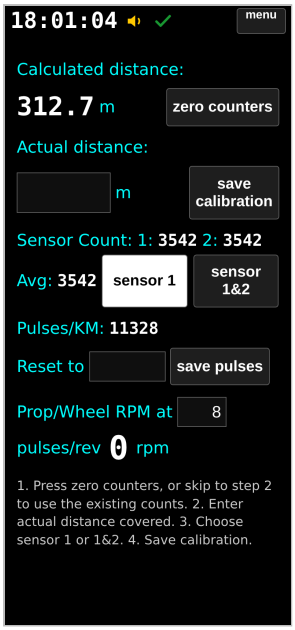

18:01:23
BEEP ASSIST & TONE
Beep navigation off on
Navigation advance 100 m
Beep timing off on
Timing advance 0 s
Phone sound lead - 50 ms +
Ahead/Behind Tone off on
Tone when behind 1 2
SYSTEM Nero2.1
Rally type italian greek surrey
Speed units kph mph
Distance units 00.00 0,000
Reaction time offset - 0.00 s +
Measure type AB,BC AB,AC
Stage data logging off on
Sensor logging
menu

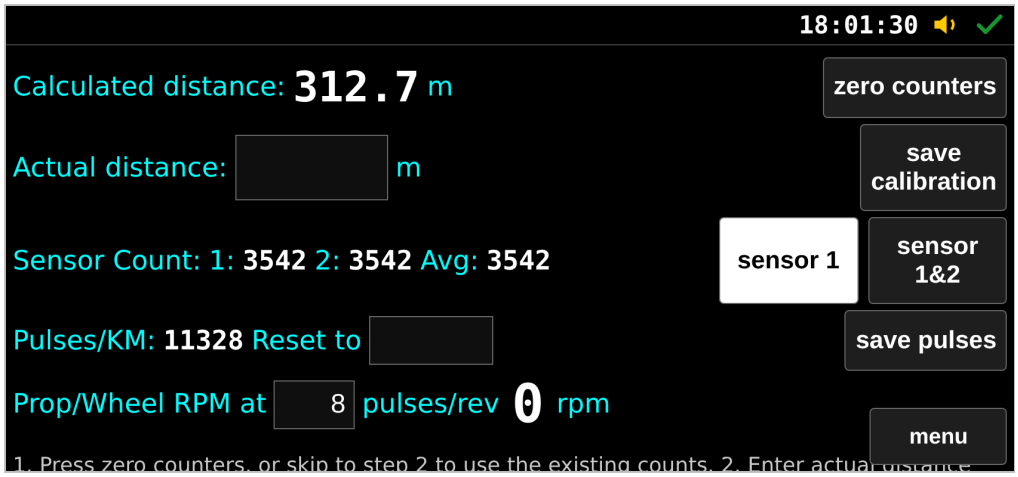
Time


18:01:11
menu
Rally time
18:01:11.1
−0.1
+0.1
Computer time
04:20:39.8
set rally to computer time
Phone time
04:20:39.8
set computer to phone time
Enter rally time
hh:mm:ss
set rally to entered time
Offset to computer
+13:40:31.3
Thu, 05 Nov 2026


18:01:36
Rally time
18:01:36.6
−0.1
+0.1
Computer time
04:21:05.3
set rally to computer time
Phone time
04:21:05.3
set computer to phone time
Enter rally time
hh:mm:ss
set rally to entered time
Offset to computer
+13:40:31.3
Thu, 05 Nov 2026
menu

Cal


18:01:04
menu
Calculated distance:
312.7 m
zero counters
Actual distance:
m
save calibration
Sensor Count: 1: 3542 2: 3542
Avg: 3542
sensor 1
sensor 1&2
Pulses/KM: 11328
Reset to
save pulses
Prop/Wheel RPM at 8
pulses/rev 0 rpm
1. Press zero counters, or skip to step 2 to use the existing counts. 2. Enter actual distance covered. 3. Choose sensor 1 or 1&2. 4. Save calibration.


18:01:30
Calculated distance: 312.7 m
zero counters
Actual distance:
m
save calibration
Sensor Count: 1: 3542 2: 3542 Avg: 3542
sensor 1
sensor 1&2
Pulses/KM: 11328 Reset to
save pulses
Prop/Wheel RPM at 8 pulses/rev 0 rpm
menu

BEEP ASSIST

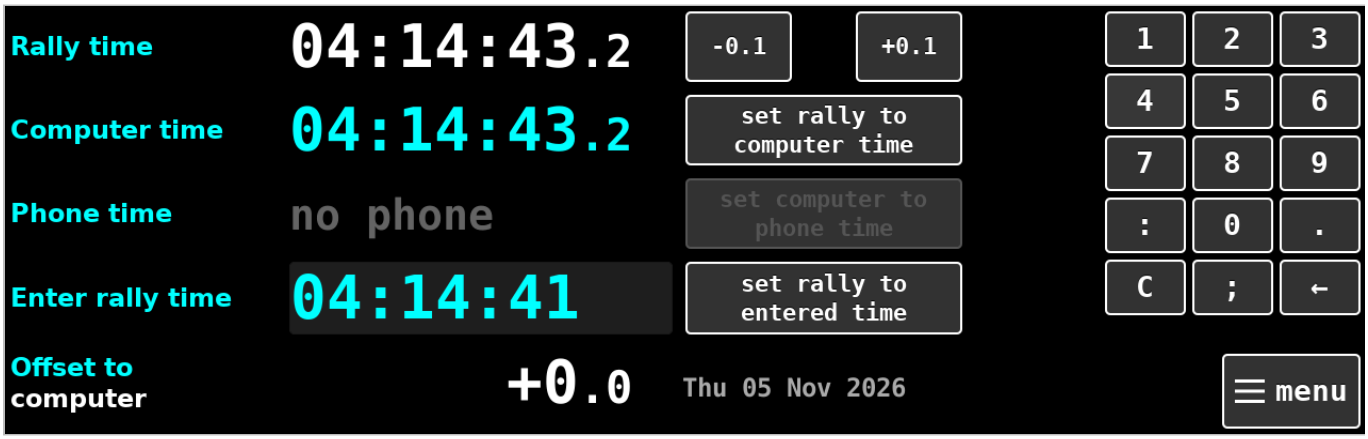
Beep Assist uses the segments **to** figures as the roadbook boxes; nothing more is entered. Switch it on in settings. Navigation double-beeps a set distance before each box. Timing single-beeps when the car should pass it, revealing distance errors from wheel spin, corner cutting or a skipping sensor. Beeps follow Total, so +10, -10 or set moves them all.

- **Beep navigation** – off/on. **Navigation advance**: metres before the box for the double beep (100 recommended).
- **Beep timing** – off/on. **Timing advance**: seconds before the box time for the single beep (0 recommended).

> In settings, Beep navigation, Beep timing and their advance figures are greyed while the rally type is italian; the choices are kept for greek. Beep Assist is for greek stages and is silent in italian. There, inside a pc, the box instead beeps the countdown to the end tube: the navigation double beep at 5 s, a single beep at 4, 3, 2 and 1, and a shorter beep on 0. The box speaker's countdown beeps sound about 0.1 s early to cover the audio delay (an estimate, not yet measured on the Pi). The same countdown sounds on a navi row, to warn that next is due. Every row that is not a pc also beeps once at 100 m before its end.

SETTING RALLY CONFIGURATIONS

DATE / TIME



The date / time page.

Rally time (white) is the clock all stages are timed against; **Computer time** (cyan) is the Raspberry Pi's clock; **Phone time** is the linked phone's, or `no phone`. **Offset to computer** is how far rally time is ahead (+) or behind (–), for example `+2.3`, or `–11:49.2` beyond a minute.

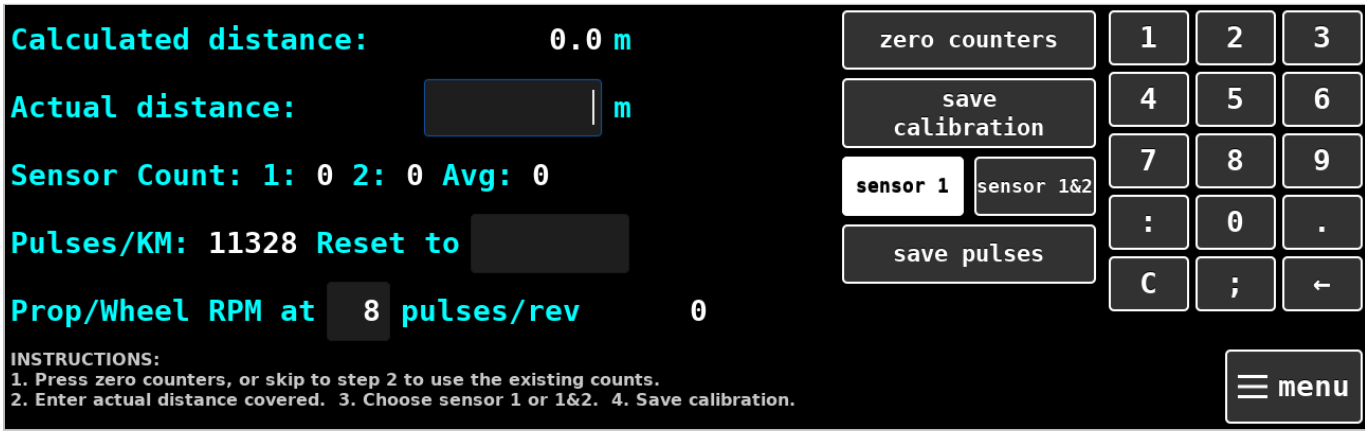
Setting the rally clock

- `set rally to computer time` makes them equal.
- Type `hh:mm:ss` (or `hh:mm`, no tenths) in **Enter rally time** and press `set rally to entered time`. Today's date is used.
- `set computer to phone time` sets the Pi's clock from the phone, leaving the rally clock as it is. Grey until a phone connects. It needs the Pi's usual password-free permission to set its clock; without it the box says `The computer clock could not be set` and nothing changes.

Trimming

`–0.1` / `+0.1` trim the rally clock by a tenth of a second; hold to repeat. Saved at once.

CALIBRATION



The calibration page.

Calibrating over a set distance

1. Press `zero counters` (or skip, to use the distance since the last Total reset).
2. Drive the measured distance (from 500 m to 100,000 m).
3. Type the **actual** distance driven (not what the box shows) in Actual distance.
4. Choose the counter: `sensor 1` or `sensor 1&2` (shown in white).
5. Press `save calibration`.

The page shows the calculated distance, each sensor's count and the pulses per km in use. If the distance is out of range or nothing was counted, `save calibration` silently does nothing, so check the pulses per km figure has changed.

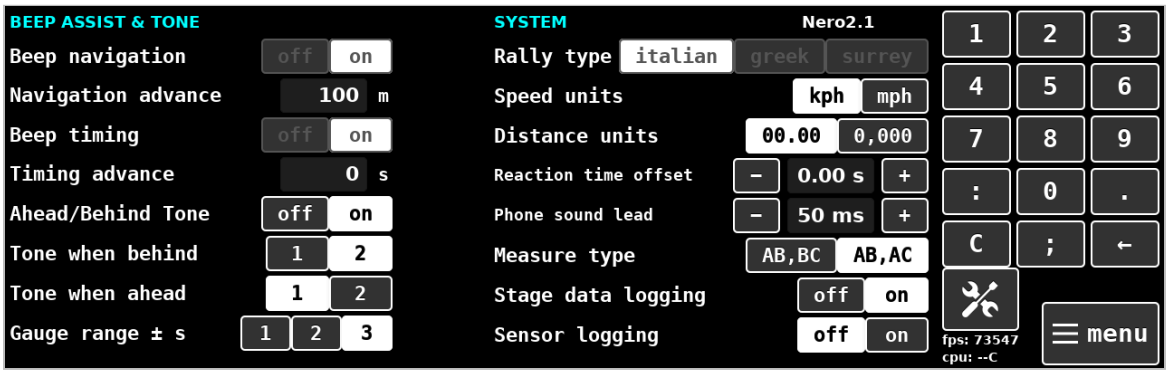
Setting the pulses directly

If the figure is known, type it in **Reset to** and press `save pulses`; no driving needed. Zero or less is ignored.

Prop/Wheel RPM

The last row shows propshaft or wheel rpm from sensor 1. Type the sensor's pulses per revolution in the small box.

SETTINGS



The settings page (italian rally type; the other types are greyed while a sector is running).

Each switch shows its present state lit. Changes are saved at once.

Tone

- **Ahead/Behind Tone** – the driver's error tone, off/on.
- **Tone when behind** / **Tone when ahead** – type 1 or 2; both silent under 0.2 s error and in the first 250 m (in italian from the moment a pc or pm starts). **Type 1**: a buzzer tick that speeds up with the error in 0.2 s steps, from about one a second at 0.2 s to about eight by 3 s; behind ticks slightly higher. **Type 2**: a steady tone rising in pitch in 0.2 s steps to 3 s; behind is higher and harsher, ahead lower and smoother.
- **Gauge range ± s** – 1, 2 or 3 seconds at each end of the gauge.

System

- **Rally type** – greek / surrey / italian. See Rally types. The software version (Nero2.1) is shown above it.
- **Reaction time offset** – −0.30 to +0.30 s in steps of 0.01 (default 0). A press at a point is taken as that much earlier, and the car that much further back. It applies to the italian tube `next`, the greek `next`, `start now` when moving at 5 kph or more, and the surrey start and finish. It does not apply to autostart, `±10`, `set` or `prev`. Positive means the crew press late. Find your figure by comparing the box's pc times with the official results.
- **Phone sound lead** – – / figure / + in steps of 10 ms, 0 to 400 (default 100 ms). It sets how early the box tells the phone to sound an Italian countdown beep, to cover the phone's audio delay. It applies to the countdown beeps only, not the 100 m beeps, and the phone shows the same figure.
- **Speed units** – kph or mph.
- **Distance units** – **00.00** (km, two decimals) or **0,000** (whole metres) for the driving figures and `add distance`. It does not flip to kilometres at 100 km. The **set** pop-ups (Set Total, Set Navi), **Distance at start** and Trip history use the same unit, and so does the phone. In km a figure is cut to the 10 m below, never rounded up, so it never reads ahead of the car; a counter that reverse takes behind its start reads negative.
- **Measure type** – **AB,BC** shows the A-B, B-C zero key on the greek and surrey navigation pages; **AB,AC** hides it.
- **Stage data logging** – one file per stage, a row a second (distance, speed, ahead/behind error), from stage start to end or abort; switched on mid-stage, it starts at once. If the file cannot be written, logging silently switches off.
- **Sensor logging** – raw counter readings, many rows a second while on, so much larger files. Both logs are deleted after 60 days (at program start) and can be downloaded on the phone's Set tab.

The spanner key (keypad, bottom left) opens Setup. Under it are the screen update rate (fps) and the Pi's CPU temperature.

SETUP



The Setup page (in the sandbox no real displays are found).

- **Screen setup** – each display found, with `rotate` and a touch keyboard switch; `reset layout`; **Force single display** puts the gauge on the co-pilot screen from the next program start.
- **Bluetooth speaker setup** – `remember playing speaker` stores the speaker now playing; `connect` / `disconnect` it later.
- **Wi-Fi setup** – one key per saved network: the box's own **hotspot** (lit when used) or **join** another. Below: the phone app address and QR code. Opened from the phone's home screen, it runs without browser bars.
- `check for software updates` – closes the program and checks for an update (needs internet).
- `settings` returns to the settings page.

INDEX

Abort stage 14, 29
Ahead / behind 14, 17, 19, 20, 21, 27, 29
Autostart 5, 7, 8, 9, 17, 20
Beep assist 5, 26
Calibration 4, 14, 28
Chevron 19, 20, 21
pc, pm, ct, co 3, 12, 13, 14, 15, 21, 26, 29
Countdown 9, 17, 18, 20, 21, 26, 29
Date and time 4, 27
Distance +10 / –10 ..... 9, 10, 15, 16, 22, 26
Driver display 19, 20, 21
Greek rally type 5, 7, 9, 17, 18, 19, 26, 29
Italian rally type 12, 13, 21, 26, 29
Late arrival flag 5, 8, 12, 22
Navi 12, 13, 14, 15, 16, 18, 21
Next speed 5, 9, 10, 18
Phone app 13, 18, 23, 27, 29, 30
Pulses per km 4, 28
Rally clock 14, 22, 27
Reaction time 29
Reset 7, 8, 15, 18, 28
Reverse 9, 10, 15, 16, 22, 29
Sector 3, 12, 13, 14, 15, 16, 21, 29
Segments 4, 5, 7, 11, 12, 13, 17, 26
Settings 4, 5, 12, 26, 29, 30
Setup 4, 29, 30
Single-display mode 22
Speed change 5, 7, 9, 10, 17, 18, 19, 20
Surrey rally type 17, 18, 19, 29
Tone 4, 29
Total 7, 9, 14, 15, 16, 17, 18, 19, 21
Trip 7, 9, 10, 17, 18, 19, 22
Units (kph / mph) 5, 7, 13, 29
Upload stage 6, 13, 17
Wi-Fi 4, 30
Zero (driver) 3, 7, 9, 10, 14, 15, 18, 28